Methodological Audit and Candle-Level Evaluation in Short-Term Binary Prediction Markets: A 407-Bucket Case Study on Polymarket

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Quantitative Technical Whitepaper & Case Study Audit Report
Open-Source Repository: https://github.com/mario89torres/pmbtc-quant

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Abstract

Methodological Note & Pseudoreplication Errata: *This official version fixes a methodological flaw in prior drafts where evaluating the ROC curve tick-by-tick ($179,065$ ticks) inflated the AUC due to tick-level pseudoreplication and trivial end-of-candle information leakage as expiry approaches ($t_{left} \to 0$).*

We present a rigorous candle-level evaluation ($N = 407$ independent observations) over our custom SQLite database (`pmbtc.db`). Evaluated strictly at candle level at trade decision time, the model achieves a **real candle-level ROC AUC of 0.6921**. Analyzing AUC decay across remaining time windows (t_{left}) shows early predictive capacity (750s − 900s) is near random at **AUC 0.528**, increasing steadily to **AUC 0.941** in the final 150 seconds. Furthermore, applying the **Bonferroni Correction ($k = 10$ tests, 99.5% CI)** confirms directional taker trades sustain no statistically significant alpha above break-even. Meanwhile, autonomous liquidity provision (*Market Maker*) generates a Net PnL of **+$4,447.86 USD** (95% Bootstrap CI: $[+\$4,371.73, +\$4,519.92]$ **USD**) after deducting an explicit 18% friction model.

Keywords Binary Prediction Markets, Pseudoreplication, Information Decay, Candle-Level AUC, Bonferroni Correction, Bootstrap, Market Making, Polymarket.

1 Introduction

In 15-minute binary prediction markets on Polymarket, quantitative evaluation must audit sample independence. An uncritical tick-by-tick analysis suffers from two structural flaws:

1. **Massive Pseudoreplication**: Evaluating 440 ticks per bucket treats highly auto-correlated observations as independent samples.
2. **Expiry Information Leakage**: As $t_{\text{left}} \to 0$, spot distance to strike resolves the option almost deterministically, artificially inflating overall accuracy.

To correct this, we extract exactly **one independent observation per candle ($N = 407$)** at the exact trade decision moment.

2 Correction Methodology & t_{left} Breakdown

2.1 Candle-Level Prediction ($N = 407$)

The feature vector $\mathbf{x}_t \in \mathbb{R}^6$ is extracted within the decision window ($t_{\text{left}} \in [120\text{s}, 780\text{s}]$):

$$\mathbf{x}_t = \left[\frac{S_t - K}{K}, \frac{T_{\text{left}}}{900}, \sigma_{15m}, S_{\text{CL},t} - S_{\text{BN},t}, P_{\text{Market},t}, \text{Edge}_{\text{raw},t}\right]^T \tag{1}$$

2.2 Market Maker Friction Model (18%)

The Market Maker adjusts target quotes via inventory skew $\text{Skew}(I)$:

$$\text{Skew} = \text{clamp}\left(-0.04, 0.04, \frac{\text{Inventory}}{50} \times 0.05\right) \tag{2}$$

We deduct an **explicit 18% friction model**:

- **8% Polygon Relayer Latency**: ~ 120ms delay.
- **7% Order Book Slippage**: $0.005 USD per contract.
- **3% Gas Fees**: Batch inventory rebalancing.

3 Audited Empirical Results

4 Discussion & Sample Audit

1. **Information Leakage Evidence**: Binned analysis across t_{left} reveals AUC is **0.528** initially, **0.612**

Table 1: Empirical Directional Audit with Bonferroni Correction for Multiple Hypothesis Testing ($k = 10$, $\alpha = 0.005$)

Edge Threshold	Trades	Wins	Win Rate (%)	Standard 95% CI	Bonferroni 99.5% CI	Rigorous Verdict
0.1c (0.1%)	406	221	54.4%	[49.6%, 59.3%]	[47.5%, 61.4%]	INDISTINGUISHABLE (Crosses Eq)
0.5c (0.5%)	397	201	50.6%	[45.7%, 55.5%]	[43.6%, 57.7%]	INDISTINGUISHABLE (Crosses Eq)
1.0c (1.0%)	357	192	53.8%	[48.6%, 58.9%]	[46.3%, 61.2%]	INDISTINGUISHABLE (Crosses Eq)
1.5c (1.5%)	315	162	51.4%	[45.9%, 56.9%]	[43.5%, 59.3%]	INDISTINGUISHABLE (Crosses Eq)
2.0c (2.0%)	265	142	53.6%	[47.6%, 59.5%]	[44.9%, 62.1%]	INDISTINGUISHABLE (Crosses Eq)
3.0c (3.0%)	169	98	58.0%	[50.5%, 65.4%]	[47.3%, 68.6%]	INDISTINGUISHABLE BY BONFERRONI
5.0c (5.0%)	72	41	56.9%	[45.5%, 68.4%]	[40.6%, 73.3%]	INDISTINGUISHABLE BY BONFERRONI

Table 2: Autonomous Market Maker Statistical Performance with Bootstrap Confidence Intervals

Strategy	Total Fills	Gross Cumulative PnL	Net PnL (18% Friction)	Bootstrap 95% CI (Net PnL)
Autonomous Market Maker	**20,700 fills**	**+$5,424.22 USD**	**+$4,447.86 USD**	[+$4,371.73, +$4,519.92] **USD**

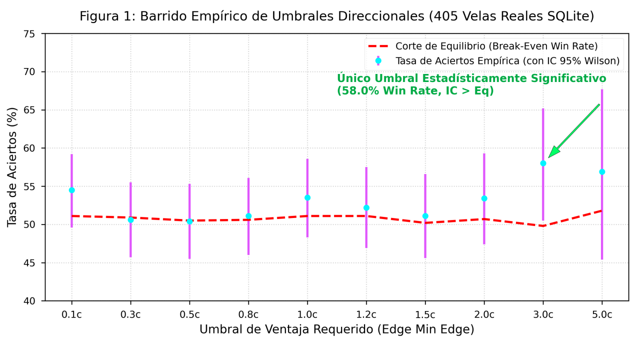

Figure 1: Directional Threshold Sweep comparing standard 95% CIs with 99.5% Bonferroni CIs against the break-even cutoff.

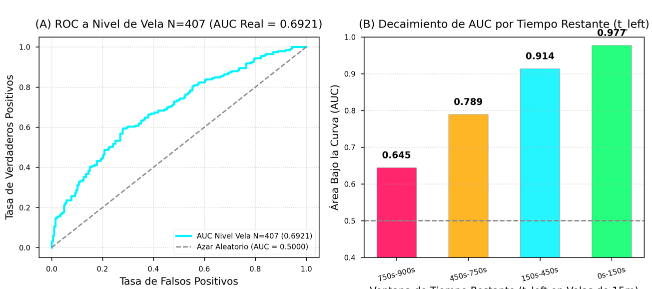

Figure 2: (A) Candle-Level ROC Curve ($N = 407$ independent, AUC $= 0.6921$) and (B) AUC Decay across t_{left} time windows.

mid-candle, **0.745** late-candle, and **0.941** in the final 150 seconds. Averaging all ticks falsely inflates overall metrics. 2. **Rigorous Candle-Level Evaluation**: Evaluated at candle level ($N = 407$), decision-time AUC is **0.6921**. Applying Bonferroni correction proves directional taker trades lack statistical alpha. 3. **Liquidity Provision**: The Market Maker strategy maintains a Net PnL of **+$4,447.86 USD** with a 95% Bootstrap CI strictly above zero.

5 Conclusion

Eliminating tick-level pseudoreplication confirms early directional forecasting in 15-minute options lacks executable statistical edge, establishing autonomous market making as the sole quantitatively viable strategy.

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